Introduction to Online Convex Optimization

Reading Seminar — Sections 5.5–5.8

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1. Randomized Regularization

1.1 Follow-the-Perturbed-Leader

Algorithm 16 — Follow-the-perturbed-leader for convex losses

1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
2: Let $x_1 = \mathbb{E}_{n\sim\mathcal{D}}[\arg\min_{x\in\mathcal{K}}\{n^\top x\}]$.
3: **for** $t = 1$ to T **do**
4: Predict x_t.
5: Observe f_t, suffer $f_t(x_t)$, and set $\nabla_t = \nabla f_t(x_t)$.
6: Update

$$x_{t+1} = \mathbb{E}_{n\sim\mathcal{D}}\left[\arg\min_{x\in\mathcal{K}}\left\{\eta\sum_{s=1}^{t}\nabla_s^\top x + n^\top x\right\}\right] \qquad (5.4)$$

7: **end for**

- Resembles a **special case of FTRL** (Follow-the-Regularized-Leader).
- **Key difference:** the regularizer is **randomized** — noise $n \sim \mathcal{D}$ is injected into the objective in place of a fixed regularizer.

Randomization is another way to make the leader **stable**: instead of adding a strongly convex regularizer, we add a random linear perturbation.

In this section the losses are chosen in the **oblivious** setting:

$$f_1, \ldots, f_T$$

are fixed ahead of time and do not depend on the learner's random choices.

Algorithm 16.

Compute the expected perturbed leader: $x_t = \mathbb{E}_n[\text{leader under } n]$.

Algorithm 17.

For linear losses, one sampled perturbation n_0 is enough in expectation.

The perturbation distribution is measured by two parameters:

$$\mathbb{E}_{n\sim\mathcal{D}}[\|n\|_a^*] = \sigma \quad \text{and} \quad \int_n |\mathcal{D}(n) - \mathcal{D}(n-u)|\,\mathrm{d}n \leq L\|u\|_a^*.$$

Size. σ controls the perturbation cost

$$\frac{1}{\eta} \cdot \sigma D$$

Sensitivity. L controls how much the leader changes when the density is shifted by u.

For the uniform distribution on $[0,1]^n$ and the Euclidean norm,

$$\sigma_2 \leq \sqrt{n}, \quad L_2 \leq 1.$$

Theorem (5.8)

Let $\mathcal{D}$ be (σ, L)-stable with respect to the norm $\|\cdot\|_a$. Then FPL (Algorithm 16) attains

$$\mathrm{Regret}_T \leq \eta D (G^*)^2 L T + \frac{1}{\eta} \sigma D,$$

and, optimizing over η,

$$\mathrm{Regret}_T \leq 2 L D G^* \sqrt{\sigma T}.$$

Definition ((σ, L)-stability)

For $\sigma, L \in \mathbb{R}$, a distribution $\mathcal{D}$ is (σ, L)-stable with respect to $\|\cdot\|_a$ if

$$\mathbb{E}_{n \sim \mathcal{D}}[\|n\|_a^*] = \sigma \quad \text{and} \quad \forall u : \int_n |\mathcal{D}(n) - \mathcal{D}(n - u)| \, \mathrm{d}n \leq L \|u\|_a^*.$$

Stability cost. The first term controls how far consecutive expected leaders can move.

Perturbation cost. The second term pays for comparing the fake initial leader with $x^\star$.

The proof repeatedly uses the FTL-BTL lemma from Section 5.2.

Lemma (5.4)

If $x_{t+1} \in \arg\min_{x \in \mathcal{K}} \sum_{s=0}^{t} g_s(x)$, then for every $u \in \mathcal{K}$,

$$\sum_{t=0}^{T} g_t(u) \geq \sum_{t=0}^{T} g_t(x_{t+1}).$$

In FPL, the perturbation is treated as a **fake initial loss**

$$g_0^n(x) = \frac{1}{\eta} n^\top x$$

This makes perturbed-leader analysis look exactly like the FTRL proof.

1.6 Proof of Theorem 5.8: Random Leaders

For fixed perturbation n, define the random leader and fake loss:

$$x_t^n = \arg\min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} \nabla_s^\top x + n^\top x \right\},$$

$$g_0^n(x) := \frac{1}{\eta} n^\top x, \quad g_t(x) := \nabla_t^\top x.$$

Applying Lemma 5.4 to $g_0^n, g_1, \dots, g_T$:

$$\begin{aligned}
\mathbb{E}_n\left[\sum_{t=0}^{T} g_t(u)\right] &\geq \mathbb{E}_n\left[g_0^n(x_1^n) + \sum_{t=1}^{T} g_t(x_{t+1}^n)\right] \\
&\geq \mathbb{E}_n[g_0^n(x_1^n)] + \sum_{t=1}^{T} g_t(\mathbb{E}_n[x_{t+1}^n]) \\
&= \mathbb{E}_n[g_0^n(x_1^n)] + \sum_{t=1}^{T} g_t(x_{t+1}).
\end{aligned}$$

Taking $u = x^\star$ in the previous slide,

$$\begin{aligned}\sum_{t=1}^{T} \nabla_t^\top (x_t - x^\star) &\le \sum_{t=1}^{T} g_t(x_t) - \sum_{t=1}^{T} g_t(x_{t+1}) + \mathbb{E}_n[g_0^n(x^\star) - g_0^n(x_1^n)] \\ &\le \sum_{t=1}^{T} \nabla_t^\top (x_t - x_{t+1}) + \frac{1}{\eta}\mathbb{E}_n[\|n\|^* \|x^\star - x_1^n\|] \\ &\le \sum_{t=1}^{T} \nabla_t^\top (x_t - x_{t+1}) + \frac{1}{\eta}\sigma D.\end{aligned}$$

Convexity of f_t and Cauchy-Schwarz give

$$\sum_t f_t(x_t) - \sum_t f_t(x^\star) \le G^* \sum_{t=1}^{T} \|x_t - x_{t+1}\| + \frac{1}{\eta}\sigma D. \quad (5.5)$$

1.8 Proof of Theorem 5.8: Density Shift

It remains to show that consecutive expected leaders are close. Define

$$h_t(n) = \arg\min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} \nabla_s^\top x + n^\top x \right\}.$$

Then $x_t = \mathbb{E}_{n[h_t(n)]}$. The next leader can be written using the same h_t but with a shifted perturbation:

$$\begin{aligned} x_t &= \int_n h_t(n) \mathcal{D}(n) \, \mathrm{d}n, \\ x_{t+1} &= \int_n h_t(n + \eta \nabla_t) \mathcal{D}(n) \, \mathrm{d}n \\ &= \int_n h_t(n) \mathcal{D}(n - \eta \nabla_t) \, \mathrm{d}n. \end{aligned}$$

Thus stability of decisions reduces to sensitivity of the density $\mathcal{D}$.

Using the density-shift expression,

$$
\begin{aligned}
\|x_t - x_{t+1}\| &= \left\| \int_n h_t(n) \cdot (\mathcal{D}(n) - \mathcal{D}(n - \eta\nabla_t)) \,\mathrm{d}n \right\| \\
&= \left\| \int_n (h_t(n) - h_t(0)) \cdot (\mathcal{D}(n) - \mathcal{D}(n - \eta\nabla_t)) \,\mathrm{d}n \right\| \\
&\leq \int_n \|h_t(n) - h_t(0)\| |\mathcal{D}(n) - \mathcal{D}(n - \eta\nabla_t)| \,\mathrm{d}n \\
&\leq D \int_n |\mathcal{D}(n) - \mathcal{D}(n - \eta\nabla_t)| \,\mathrm{d}n \\
&\leq DL\eta\|\nabla_t\|^* \leq \eta DLG^*.
\end{aligned}
$$

The only real input here is (σ, L)-stability with $u = \eta\nabla_t$.

1.10 Proof of Theorem 5.8: Final Bound

Substitute the shift bound into (5.5):

$$\begin{aligned}
\mathrm{Regret}_T &\le G^* \sum_{t=1}^{T} \left\| x_t - x_{t+1} \right\| + \frac{1}{\eta}\sigma D \\
&\le G^* \sum_{t=1}^{T} \eta D L G^* + \frac{1}{\eta}\sigma D \\
&= \eta D L (G^*)^2 T + \frac{1}{\eta}\sigma D.
\end{aligned}$$

Optimizing over η gives the stated theorem bound:

$$\mathrm{Regret}_T \le 2LDG^* \sqrt{\sigma T}.$$

For uniform noise on $[0, 1]^n$, this gives a loose $O\left(DGn^{\frac{1}{4}}\sqrt{T}\right)$ bound.

Algorithm 17 — FPL for linear losses

1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
2: Sample $n_0 \sim \mathcal{D}$. Let $\hat{x}_1 \in \arg\min_{x \in \mathcal{K}} \{-n_0^\top x\}$.
3: **for** $t = 1$ to T **do**
4: Predict $\hat{x}_t$.
5: Observe the linear loss function, suffer loss $g_t^\top \hat{x}_t$.
6: Update

$$\hat{x}_t = \arg\min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} g_s^\top x + n_0^\top x \right\}$$

7: **end for**

Linear loss. The exact expectation in Algorithm 16 can be replaced by one perturbation sample.

Expected regret. The guarantee is taken over the sampled perturbation.

1.12 Why One Sample is Enough for Linear Losses

Let

$$w_t(n) = \arg\min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t} g_s^\top x + n^\top x \right\}.$$

Algorithm 16 predicts $x_t = \mathbb{E}_n[w_{t-1}(n)]$.

For a linear loss $f_t(x) = g_t^\top x$,

$$f_t(x_t) = f_t(\mathbb{E}_n[w_{t-1}(n)]) = \mathbb{E}_n[f_t(w_{t-1}(n))].$$

Therefore sampling one $n_0 \sim \mathcal{D}$ and playing $\hat{x}_t = w_{t-1}(n_0)$ has the same expected loss as the exact expectation-based prediction.

Theorem (Corollary 5.9)

For Algorithm 17,

$$\mathbb{E}_{n_0\sim\mathcal{D}}\left[\sum_{t=1}^{T} f_t(\hat{x}_t) - \sum_{t=1}^{T} f_t(x^\star)\right] \le \eta L D (G^*)^2 T + \frac{1}{\eta}\sigma D.$$

Computational point. Algorithm 17 only needs a linear optimization oracle over $\mathcal{K}$; the set need not even be convex for the oracle call itself.

1.14 FPL for Prediction from Expert Advice

Algorithm 18 — FPL for prediction from expert advice

Input: $\eta > 0$.
Draw n exponentially distributed variables $n(i) \sim e^{-\eta x}$.
Let $x_1 = \arg\min_{e_i \in \Delta_n} \{-e_i^\top n\}$.
for $t = 1$ to T **do**
  Predict using expert i_t such that $\hat{x}_t = e_{i_t}$.
  Observe the loss vector, suffer loss $g_t^\top \hat{x}_t = g_t(i_t)$.
  Update (w.l.o.g. choose $\hat{x}_{t+1}$ to be a vertex)

$$\hat{x}_{t+1} = \arg\min_{x \in \Delta_n} \left\{ \sum_{s=1}^{t} g_s^\top x - n^\top x \right\}$$

end for

- Special case of Algorithm 17: the simplex Δ_n, costs in $[0,1]$, and one-sided exponential noise $n(i) \sim e^{-\eta x}$ — the first use of perturbation in decision making (Hannan [1957]).
- The generic FPL bound (Corollary 5.9) is loose here, so Theorem 5.10 gives a dedicated analysis, tight up to constants.

1.15 Exponential Noise for Expert Advice

For expert advice, the perturbation is one-sided exponential noise:

$$\Pr[n(i) \leq x] = 1 - e^{-\eta x} \quad (x \geq 0).$$

The update chooses the expert with smallest cumulative loss after subtracting its random bonus:

$$\hat{x}_{t+1} = \arg\min_{x \in \Delta_n} \left\{ \sum_{s=1}^{t} g_s^\top x - n^\top x \right\}.$$

Corollary 5.9 is too loose for this simplex/exponential-noise case, so Theorem 5.10 uses a direct leader-change analysis.

Theorem (5.10)

Algorithm 18 outputs predictions $\hat{x}_1, \ldots, \hat{x}_T \in \Delta_n$ with

$$(1-\eta)\mathbb{E}\left[\sum_t g_t^\top \hat{x}_t\right] \le \min_{x^\star \in \Delta_n} \sum_t g_t^\top x^\star + \frac{4 \log n}{\eta}.$$

Optimizing η. Choosing $\eta = \sqrt{(\log n)/T}$ gives

$$\mathrm{Regret}_T = O\left(\sqrt{T \log n}\right),$$

matching the **Hedge** guarantee (Theorem 1.5) up to constants.

Why a separate analysis? — the order gap.

- **Generic FPL (Cor. 5.9)** on Δ_n: the uniform-noise instance has $\sigma \le \sqrt{n}$, $L \le 1$, giving $\mathrm{Regret}_T = O\left(DGn^{1/4}\sqrt{T}\right)$ — a factor $n^{1/4}$ **worse** than the $O\left(\sqrt{T}\right)$ of OGD (Theorem 3.1).
- **Expert-advice FPL (Thm 5.10)**, with exponential noise, removes that polynomial-in-n penalty: only a $\sqrt{\log n}$ dependence remains, on par with the standard multiplicative-weights / Hedge bound $O(\sqrt{T \log n})$.

1.17 Proof of Theorem 5.10 (1/2)

Proof. Set $g_0 = -n$. Applying **Lemma 5.4** (FTL-BTL) to $f_t(x) = g_t^\top x$,

$$\mathbb{E}\left[\sum_{t=0}^{T} g_t^\top u\right] \geq \mathbb{E}\left[\sum_{t=0}^{T} g_t^\top \hat{x}_{t+1}\right].$$

Therefore, telescoping and isolating the $t = 0$ term,

$$\begin{aligned}\mathbb{E}\left[\sum_{t=1}^{T} g_t^\top (\hat{x}_t - x^\star)\right] &\leq \mathbb{E}\left[\sum_{t=1}^{T} g_t^\top (\hat{x}_t - \hat{x}_{t+1})\right] + \mathbb{E}[g_0^\top (x^\star - x_1)] \\ &\leq \mathbb{E}\left[\sum_{t=1}^{T} g_t^\top (\hat{x}_t - \hat{x}_{t+1})\right] + \mathbb{E}\left[\|n\|_\infty \|x^\star - x_1\|_1\right] \\ &\leq \sum_{t=1}^{T} \mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] + \frac{4}{\eta} \log n. \qquad (5.6)\end{aligned}$$

The 2nd line uses the **generalized Cauchy–Schwarz** inequality; the 3rd uses

$$\mathbb{E}_{n\sim\mathcal{D}}[\|n\|_\infty] \leq \frac{2\log n}{\eta} \quad \text{(exercise)}, \quad \|x^\star - x_1\|_1 \leq 2.$$

Per-step term. Bounding by the probability that the leader changes, times $\|g_t\|_\infty \le 1$ (losses bounded by one):

$$\mathbb{E}[g_t^\top(\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] \le \|g_t\|_\infty \cdot \Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t] \le \Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t].$$

$\hat{x}_t = e_{i_t}$ leads iff $-n(i_t) > v$ for some v fixed by the history; it **stays** the leader iff $-n(i_t) > v + g_t(i_t)$. By the **memorylessness** of the exponential distribution,

$$\begin{aligned}\Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t] &= 1 - \Pr[-n(i_t) > v + g_t(i_t) \mid -n(i_t) > v] \\ &= 1 - \frac{\int_{v+g_t(i_t)}^\infty \eta e^{-\eta x}\,\mathrm{d}x}{\int_v^\infty \eta e^{-\eta x}\,\mathrm{d}x} = 1 - e^{-\eta g_t(i_t)} \le \eta g_t(i_t) = \eta g_t^\top \hat{x}_t.\end{aligned}$$

Substituting into (5.6),

$$\mathbb{E}\left[\sum_{t=1}^T g_t^\top(\hat{x}_t - x^\star)\right] \le \eta \sum_t \mathbb{E}_t[g_t^\top \hat{x}_t] + \frac{4\log n}{\eta},$$

which rearranges to the theorem. □

Per-step term. Bounding by the probability that the leader changes, times $\|g_t\|_\infty \le 1$ (losses bounded by one):

$$\mathbb{E}[g_t^\top(\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] \le \|g_t\|_\infty \cdot \Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t] \le \Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t].$$

$\hat{x}_t = e_{i_t}$ leads iff $-n(i_t) > v$ for some v fixed by the history; it **stays** the leader iff $-n(i_t) > v + g_t(i_t)$. By the **memorylessness** of the exponential distribution,

$$\begin{aligned}\Pr[\hat{x}_t \ne \hat{x}_{t+1} \mid \hat{x}_t] &\le 1 - \Pr[-n(i_t) > v + g_t(i_t) \mid -n(i_t) > v] \\ &= 1 - \frac{\int_{v+g_t(i_t)}^{\infty} \eta e^{-\eta x}\,\mathrm{d}x}{\int_v^{\infty} \eta e^{-\eta x}\,\mathrm{d}x} = 1 - e^{-\eta g_t(i_t)} \le \eta g_t(i_t) = \eta g_t^\top \hat{x}_t.\end{aligned}$$

My reading: this first step looks like it should really be "$\le$", not "$=$" — I sketch why on the next slide.

Substituting into (5.6),

$$\mathbb{E}\left[\sum_{t=1}^{T} g_t^\top(\hat{x}_t - x^\star)\right] \le \eta \sum_t \mathbb{E}_t[g_t^\top \hat{x}_t] + \frac{4\log n}{\eta},$$

which rearranges to the theorem. $\square$

1.19 The Marked Equality is Only "$\leq$": an $N = 2$ Check

The proof's event is a **sufficient** condition for staying leader, not a necessary one: it ignores that competitors also pay losses.

Expert	Before g_t	After $g_{t(1)} = g_{t(2)} = 1$
1 = leader	S_1	$S_1 + 1$
2	$S_2 > S_1$	$S_2 + 1 > S_1 + 1$

Equal losses shift both scores by the same amount, so the arg min cannot change:

$$\Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t = e_1] = 0.$$

But the displayed equality would give

$$1 - e^{-\eta g_{t(1)}} = 1 - e^{-\eta} > 0.$$

Thus the marked step should be "$\leq$", not "$=$". The direction is still enough for the regret bound, so Theorem 5.10 is unaffected.

2. Adaptive Gradient Descent (AdaGrad)

2.1 Learning the Regularizer Online

Motivation. RFTL's regret bound (5.7) depends on the regularizer R; the optimal R depends on both $\mathcal{K}$ and the cost sequence. So we learn it online — AdaGrad optimizes the regularization (line 5) to shrink the dominant gradient-norm term in (5.7).

Algorithm 19 — AdaGrad

1: Input: parameters η, $x_1 \in \mathcal{K}$.
2: Initialize: $G_0 = 0$.
3: **for** $t = 1$ to T **do**
4: Predict x_t, suffer loss $f_t(x_t)$.
5: Update $G_t = G_{t-1} + \nabla_t \nabla_t^\top$ and define

$$\text{(diagonal)} \quad H_t = \arg \min_{H \succeq 0,\ H = \operatorname{diag}(H)} \{G_t \cdot H^{-1} + \operatorname{Tr}(H)\} = (\operatorname{diag}(G_t))^{\frac{1}{2}}$$

$$\text{(full matrix)} \quad H_t = \arg\min_{H \succeq 0} \{G_t \cdot H^{-1} + \operatorname{Tr}(H)\} = G_t^{\frac{1}{2}}$$

6: Update $y_{t+1} = x_t - \eta H_t^{-1} \nabla_t, \quad x_{t+1} = \arg\min_{x \in \mathcal{K}} \left\| y_{t+1} - x \right\|_{H_t}^2.$
7: **end for**

AdaGrad chooses the matrix H_t that would have made the **observed** gradients small, with a trace penalty to keep the metric from blowing up:

$$H_t = \arg\min_H \{G_t \cdot H^{-1} + \mathrm{Tr}(H)\}, \quad G_t = \sum_{s=1}^{t} \nabla_s \nabla_s^\top .$$

Definition (Two comparator classes)

$$\mathcal{H}_1 = \{H : H = \mathrm{diag}(H), H \succeq 0, \mathrm{Tr}(H) \leq 1\}, \quad \mathcal{H}_2 = \{H : H \succeq 0, \mathrm{Tr}(H) \leq 1\}.$$

Lemma (5.11)

For $i \in \{1, 2\}$, with the corresponding final AdaGrad matrix H_T,

$$\sqrt{\min_{H \in \mathcal{H}_i} \sum_{t=1}^{T} \|\nabla_t\|_H^{*2}} = \mathrm{Tr}(H_T).$$

2.3 Regret Bound: Compete with the Best Fixed Metric

Theorem (5.12)

Let x_t be generated by Algorithm 19. With $\eta = \frac{D_\infty}{\sqrt{2}}$ for the diagonal version and $\eta = \frac{D}{\sqrt{2}}$ for the full-matrix version, for any $x^\star \in \mathcal{K}$,

$$\mathrm{Regret}_{T(\text{AdaGrad-diag})} \le \sqrt{2} D_\infty \sqrt{\min_{H \in \mathcal{H}_1} \sum_t \|\nabla_t\|_H^{*2}},$$

$$\mathrm{Regret}_{T(\text{AdaGrad-full})} \le \sqrt{2} D \sqrt{\min_{H \in \mathcal{H}_2} \sum_t \|\nabla_t\|_H^{*2}}.$$

Reading. AdaGrad is nearly as good as the best fixed quadratic regularizer in hindsight, chosen from the diagonal or full PSD trace ball.

Why the theorem follows from Lemma 5.11.

The adaptive choice makes $\mathrm{Tr}(H_T)$ exactly encode the best hindsight gradient norm over $\mathcal{H}_i$.

Why there is still a price.

The regularizer changes over time, so the proof needs one extra drift term involving $H_t - H_{t-1}$.

2.4 When AdaGrad Beats OGD

For the unit cube in $\mathbb{R}^d$, $D_\infty = 1$ while $D = \sqrt{d}$. The textbook compares the diagonal-AdaGrad and OGD bounds as

$$\mathrm{Regret}_{T(\text{AdaGrad-diag})} \le \sqrt{2\ \mathrm{Tr}\Big((\mathrm{diag}(G_T))^{\frac{1}{2}}\Big)},$$

$$\mathrm{Regret}_{T(\text{OGD})} \le \sqrt{2d\ \mathrm{Tr}(\mathrm{diag}(G_T))}.$$

Sparse gradient geometry.

If only a few coordinates accumulate large gradient mass, AdaGrad can improve over OGD by up to a $\sqrt{d}$ factor.

Dense / Euclidean geometry.

If G_T is dense, or the feasible set is closer to a Euclidean ball, OGD can be better by the same order.

Takeaway. AdaGrad is not uniformly better than OGD; it wins when the data geometry is coordinate-sparse or otherwise mismatched to the Euclidean metric.

The proof splits AdaGrad regret into two terms, then controls each by the same final quantity $\mathrm{Tr}(H_T)$.

Lemma (5.13)

$$\mathrm{Regret}_T \le \frac{\eta}{2}\left(G_T \cdot H_T^{-1} + \mathrm{Tr}(H_T)\right) + \frac{1}{2\eta}\sum_{t=0}^{T} \|x_t - x^\star\|^2_{H_t - H_{t-1}}.$$

Lemma 5.14. Gradient term:

$$G_T \cdot H_T^{-1} \le \mathrm{Tr}(H_T).$$

Lemma 5.15. Drift term:

$$\sum_t \|x_t - x^\star\|^2_{H_t - H_{t-1}} \le D_*^2 \ \mathrm{Tr}(H_T).$$

Choosing η balances the two terms and recovers Theorem 5.12.

2.6 Lemma 5.13: One-Step Identity

The update is a gradient step in the current metric:

$$y_{t+1} = x_t - \eta H_t^{-1} \nabla_t.$$

Subtract $x^\star$ and multiply by H_t:

$$\begin{aligned} y_{t+1} - x^\star &= x_t - x^\star - \eta H_t^{-1} \nabla_t, \\ H_t(y_{t+1} - x^\star) &= H_t(x_t - x^\star) - \eta \nabla_t. \end{aligned}$$

Multiplying the transpose of the first line by the second line gives the basic energy identity.

2.7 Lemma 5.13: Expanding the Square

Expanding in the H_t-norm gives the textbook equation (5.8):

$$\left\|y_{t+1} - x^\star\right\|_{H_t}^2 = \|x_t - x^\star\|_{H_t}^2 - 2\eta\nabla_t^\top(x_t - x^\star) + \eta^2\nabla_t^\top H_t^{-1}\nabla_t. \quad (5.8)$$

Rearranging this identity will isolate the one-step regret term $\nabla_t^\top(x_t - x^\star)$.

Since x_{t+1} is the projection of y_{t+1} in the norm induced by H_t,

$$\left\|y_{t+1} - x^\star\right\|_{H_t}^2 \geq \left\|x_{t+1} - x^\star\right\|_{H_t}^2.$$

Combining this with (5.8):

$$\nabla_t^\top (x_t - x^\star) \leq \frac{\eta}{2} \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta}\left(\left\|x_t - x^\star\right\|_{H_t}^2 - \left\|x_{t+1} - x^\star\right\|_{H_t}^2\right).$$

This is the OGD proof, except that the metric changes with t.

2.9 Lemma 5.13: Telescoping with Changing Metrics

Summing the one-step inequality from $t = 1$ to T:

$$\sum_{t=1}^{T} \nabla_t^\top (x_t - x^\star) \leq \frac{\eta}{2} \sum_{t=1}^{T} \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta} \|x_1 - x^\star\|_{H_0}^2$$
$$+ \frac{1}{2\eta} \sum_{t=1}^{T} \left(\|x_t - x^\star\|_{H_t}^2 - \|x_t - x^\star\|_{H_{t-1}}^2 \right)$$
$$- \frac{1}{2\eta} \|x_{T+1} - x^\star\|_{H_T}^2 .$$

With $H_0 = 0$ and dropping the final nonnegative term,

$$\sum_t \nabla_t^\top (x_t - x^\star) \leq \frac{\eta}{2} \sum_t \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta} \sum_{t=0}^{T} \|x_t - x^\star\|_{H_t - H_{t-1}}^2 .$$

2.10 Lemma 5.13: BTL Controls the Gradient Term

Define

$$\Psi_{t(H)} = \nabla_t \nabla_t^\top \cdot H^{-1}, \quad \Psi_0(H) = \mathrm{Tr}(H).$$

By construction, H_t minimizes $\sum_{i=0}^{t} \Psi_{i(H)}$. Applying the BTL Lemma 5.4,

$$\begin{aligned}\sum_{t=1}^{T} \nabla_t^\top H_t^{-1} \nabla_t &= \sum_{t=1}^{T} \Psi_{t(H_t)} \\ &\le \sum_{t=1}^{T} \Psi_{t(H_T)} + \Psi_0(H_T) - \Psi_0(H_0) \\ &= G_T \cdot H_T^{-1} + \mathrm{Tr}(H_T).\end{aligned}$$

Substituting this into the previous slide proves Lemma 5.13.

2.11 Lemma 5.14: Explicit Optimizer

Proposition 5.16 solves the optimization in line 5. For $A \succeq 0$,

$$\arg\min_{X \succeq 0} \{A \cdot X^{-1} + \mathrm{Tr}(X)\} = A^{\frac{1}{2}}.$$

With the diagonal constraint, the optimizer is

$$\arg \min_{X \succeq 0,\ X = \mathrm{diag}(X)} \{A \cdot X^{-1} + \mathrm{Tr}(X)\} = (\mathrm{diag}(A))^{\frac{1}{2}}.$$

Applying this to $A = G_T$ gives

$$H_T = G_T^{\frac{1}{2}} \quad \text{or} \quad H_T = (\mathrm{diag}(G_T))^{\frac{1}{2}}.$$

For the full-matrix version,

$$G_T \cdot H_T^{-1} = G_T \cdot G_T^{-\frac{1}{2}} = \mathrm{Tr}\left(G_T^{\frac{1}{2}}\right) = \mathrm{Tr}(H_T).$$

For the diagonal version, H_T ignores off-diagonal entries:

$$G_T \cdot H_T^{-1} = \mathrm{diag}(G_T) \cdot (\mathrm{diag}(G_T))^{-\frac{1}{2}} = \mathrm{Tr}\left((\mathrm{diag}(G_T))^{\frac{1}{2}}\right) = \mathrm{Tr}(H_T).$$

Therefore, in both cases,

$$G_T \cdot H_T^{-1} \leq \mathrm{Tr}(H_T).$$

2.13 Lemma 5.15: Diagonal Drift Term

For diagonal AdaGrad, $G_t \succeq G_{t-1}$ implies $H_t - H_{t-1} \succeq 0$. Since for diagonal H, $x^\top H x \leq \|x\|_\infty^2 \operatorname{Tr}(H)$,

$$\begin{aligned}
\sum_{t=1}^{T} \|x_t - x^\star\|_{H_t - H_{t-1}}^2 &\leq \sum_{t=1}^{T} D_\infty^2 \operatorname{Tr}(H_t - H_{t-1}) \\
&= D_\infty^2 \sum_{t=1}^{T} (\operatorname{Tr}(H_t) - \operatorname{Tr}(H_{t-1})) \\
&\leq D_\infty^2 \operatorname{Tr}(H_T).
\end{aligned}$$

The trace telescopes because the metric increments are PSD.

2.14 Lemma 5.15: Full-Matrix Drift Term

In the full-matrix case, $H_t - H_{t-1} \succeq 0$ and $\lambda_{\max(A)} \leq \operatorname{Tr}(A)$ for $A \succeq 0$:

$$\begin{aligned}\sum_{t=1}^{T} \|x_t - x^\star\|^2_{H_t - H_{t-1}} &\leq \sum_{t=1}^{T} D^2 \lambda_{\max(H_t - H_{t-1})} \\ &\leq D^2 \sum_{t=1}^{T} \operatorname{Tr}(H_t - H_{t-1}) \\ &= D^2 \sum_{t=1}^{T} (\operatorname{Tr}(H_t) - \operatorname{Tr}(H_{t-1})) \\ &\leq D^2 \operatorname{Tr}(H_T).\end{aligned}$$

This proves Lemma 5.15.

2.15 Putting the Pieces Together

Lemma 5.13 plus Lemmas 5.14 and 5.15 gives

$$\begin{aligned}\mathrm{Regret}_T &\leq \frac{\eta}{2}(2\ \mathrm{Tr}(H_T)) + \frac{D_*^2}{2\eta}\ \mathrm{Tr}(H_T) \\ &= \left(\eta + \frac{D_*^2}{2\eta}\right)\ \mathrm{Tr}(H_T),\end{aligned}$$

where $D_* = D_\infty$ for diagonal AdaGrad and $D_* = D$ for full AdaGrad.

Choosing $\eta = \frac{D_*}{\sqrt{2}}$ yields

$$\mathrm{Regret}_T \leq \sqrt{2} D_*\ \mathrm{Tr}(H_T).$$

Finally, Lemma 5.11 converts $\mathrm{Tr}(H_T)$ back to the best hindsight metric:

$$\mathrm{Tr}(H_T) = \sqrt{\min_{H \in \mathcal{H}_i} \sum_{t=1}^{T} \|\nabla_t\|_H^{*2}},$$

which is exactly Theorem 5.12.

3. Bibliographic Remarks

- **Origins.** Regularization for online learning: Grove et al. [2001]; Kivinen & Warmuth [2001].
- **Follow-the-(Perturbed-)Leader.** Kalai & Vempala [2005] coined *follow-the-leader* and analyzed random perturbation as a regularizer (FPL), reviving an idea of Hannan [1957].
- **Follow-the-Regularized-Leader.** *FTRL* coined by Shalev-Shwartz & Singer [2007]; identical *RFTL* by Abernethy et al. [2008]. Hazan & Kale [2008]: RFTL $\equiv$ Online Mirror Descent.
- **AdaGrad.** Duchi et al. [2010, 2011]; diagonal version in parallel by McMahan & Streeter [2010]. Analysis here follows Gupta et al. [2017].
- **Adaptive regularization in deep learning.** AdaGrad, RMSprop [Tieleman & Hinton 2012], Adam [Kingma & Ba 2014]; survey: Goodfellow et al. [2016].
- **Randomization $\leftrightarrow$ regularization.** In special cases, adding randomization $\approx$ deterministic strongly convex regularization [Abernethy et al. 2014, 2016].

4. Exercises

Claim. Let $A \succ 0$ and define $\|x\|_A := \sqrt{x^\top A x}$. Then its dual norm is

$$\|y\|_A^* = \|y\|_{A^{-1}} = \sqrt{y^\top A^{-1} y}.$$

Proof. By definition,

$$\|y\|_A^* = \sup_{\|x\|_A \le 1} x^\top y.$$

Put $z = A^{\frac{1}{2}} x$, so $x = A^{-\frac{1}{2}} z$ and $\|x\|_A = \|z\|_2$. Hence

$$\begin{aligned} \|y\|_A^* &= \sup_{\|z\|_2 \le 1} z^\top A^{-\frac{1}{2}} y \\ &= \left\| A^{-\frac{1}{2}} y \right\|_2 = \sqrt{y^\top A^{-1} y}. \end{aligned}$$

The supremum is attained in the direction $z = A^{-\frac{1}{2}} y / \left\| A^{-\frac{1}{2}} y \right\|_2$ when $y \neq 0$. $\square$

4.2 Exercise 1(b): Generalized Cauchy–Schwarz

Claim. For any norm $\|\cdot\|$ with dual norm $\|y\|_* := \sup_{\|z\|\leq 1} z^\top y$,

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

Proof. If $x = 0$ (or $y = 0$), both sides equal 0 and the inequality holds. So assume $x \neq 0$ from here on.

Since $x \neq 0$, the unit vector $u = x/\|x\|$ satisfies $\|u\| = 1$, hence it is admissible in the supremum defining the dual norm. Therefore

$$\|y\|_* = \sup_{\|z\|\leq 1} z^\top y \geq u^\top y = \frac{x^\top y}{\|x\|}.$$

Multiplying both sides by $\|x\| > 0$ yields

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

□